# JANITH GAYASHAN

AI/ML Engineer

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## PROFESSIONAL SUMMARY

B.Sc. (Hons) in Artificial Intelligence candidate at the University of Moratuwa, with a strong technical foundation in machine learning, deep learning, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and agentic AI systems. Experienced in integrating predictive models, explainable AI, retrieval systems, and autonomous agent workflows into practical AI applications. Completed an AI/ML internship at Sri Lanka Telecom, gaining industry experience in developing applied AI solutions.

## WORK EXPERIENCE

**AI/ML Intern – Sri Lanka Telecom (SLT)** March 2025 – September 2025

- Designed and implemented the agentic AI architecture for the SL-Travel tourism platform using LangGraph, coordinating specialized agents for travel planning, train and hotel booking, destination recommendations, tour-guide matching, and sentiment analysis.
- Built the platform's Retrieval-Augmented Generation (RAG) pipeline using Pinecone-based semantic retrieval to deliver grounded travel information, while integrating structured data sources for personalized destination and tour-guide recommendations.
- Integrated the multi-agent AI system with FastAPI, MySQL, MongoDB, and the Next.js frontend to enable end-to-end conversational travel workflows.

## EDUCATION

**B.Sc. (Hons) in Artificial Intelligence** 2022 – 2026

University of Moratuwa

## TECHNICAL SKILLS

**Programming:** Python, SQL, Java, C

**Machine Learning & Deep Learning:** PyTorch, Scikit-learn, LightGBM, XGBoost

**Generative AI & Agentic AI:** LangGraph, LlamaIndex, LangChain, LLMs, BERT, Sentence Transformers, BM25

**Data Analysis & Visualization:** Pandas, NumPy, Matplotlib, Seaborn

**MLOps & Deployment:** MLflow, FastAPI, Docker

**Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, Pinecone, Qdrant, Redis

**Development Tools:** Git, GitHub

## PROJECTS

**Explainable Deep Learning for Autonomous Driving (Final Year Research Project)** GitHub

- Designed an intrinsically explainable deep learning architecture for autonomous driving using a spatial attention mechanism with a ResNet-18 backbone, enabling the model to focus on driving-relevant visual regions during steering prediction.
- Evaluated the architecture across multiple random seeds against PilotNet, ResNet-18, and attention-based baselines, demonstrating improved MAE and RMSE performance while achieving a 6.6× higher Deletion AUC than SoftAttention, indicating substantially stronger explanation faithfulness.
- **Technologies:** PyTorch, OpenCV, ResNet-18, CNNs, Explainable AI (XAI)

**ERPNext Agentic AI Automation Platform – Ongoing**

- Designing and implementing an agentic AI automation platform for ERPNext to support workflows across procurement, sales, inventory, finance, HR, and business operations.
- Building LangGraph workflows with state persistence, checkpointing, human-in-the-loop approvals, dynamic routing, recovery mechanisms, ERP tool integration, and controlled LLM-based decision-making with business rules and permission checks.
- **Technologies:** LangGraph, ERPNext/Frappe, LLMs, FastAPI, MariaDB, Pinecone, Redis, Docker

**RAG-Based Insurance Document System** GitHub

- Developing an insurance document intelligence system that processes policy documents through document ingestion, semantic chunking, and structured retrieval workflows.
- Implementing hybrid BM25 and vector retrieval with query transformation, reranking, citation-grounded generation, and retrieval/generation evaluation to improve answer relevance and grounding.
- **Technologies:** LlamaIndex, Qdrant, BM25, Sentence Transformers, RAGAS, FastAPI, PostgreSQL, Docker

**Financial Transaction Fraud Detection System** GitHub

- Developing an end-to-end fraud detection system with time-aware validation, temporal feature engineering, and techniques for highly imbalanced transaction data.
- Evaluating LightGBM and XGBoost models with cost-sensitive metrics, probability calibration, threshold optimization, and SHAP-based explainability to improve fraud detection performance and interpretability.
- **Technologies:** Pandas, NumPy, Scikit-learn, LightGBM, XGBoost, SHAP, MLflow, FastAPI

**Hybrid Movie Recommender System** GitHub

- Developed a hybrid movie recommendation system on the MovieLens 1M dataset by combining collaborative filtering, content-based filtering, and popularity-based recommendation approaches.
- Implemented cold-start handling and time-aware evaluation to generate Top-N recommendations and rating predictions under realistic recommendation scenarios.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Scikit-surprise, SciPy, Matplotlib, Seaborn

## CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification – DeepLearning.AI / Coursera
- Advanced Learning Algorithms – DeepLearning.AI / Coursera
- Prompt Engineering with Llama 2 and 3 – DeepLearning.AI

## REFERENCES

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